

BE 501: Anatomy and Physiology

Course Code	BE 501
Course Name	Anatomy and Physiology
L-T-P-C	3-0-0-3
Prerequisites	IC136 / BE 201/ BE 202 or equivalent
Intended for	B. Tech M.Tech Integrated Dual Degree Bioengineering students
Distribution	Core for IDD BE, Elective for other B.Tech students
Approval	44th BoA (Nov 2021)

Course Contents

- **Foundations of Anatomy, Physiology and homeostasis**
Organization of the Human Body, Chemical Foundations –Atoms, Ions, Molecules, Bonds, Solutions comprising different specialized tissues, organs and organ systems. [4 Lectures]
- **Haemopoietic system, Lymphatic System and Endocrine system**
Composition and functions of blood and its elements, their disorders, blood groups (ABO classification) and their significance, mechanism of coagulation, Anaemias and its types, lymph organs. Anatomy and physiology of Pituitary, thyroid, parathyroid, adrenal and pancreatic glands, specific hormones and disorders of these glands, endocrine control of growth and metabolism; pineal, thymus. [8 Lectures]
- **Cardiovascular and Musculo-skeletal system**
Anatomy and physiology of the heart, cardiac cycle; circulation of blood, heart rate, blood pressure, ECG and heart sounds, lymphatic vessel, systemic and portal circulation; vascular system –arteries, arterioles, capillaries, venules. Blood pressure and its regulation. Brief outline of cardiovascular disorders like hypertension, myocardial infarction, congestive heart failure, cardiomyopathies and cardiac arrhythmias. Anatomy and physiology of muscular system, types of muscle tissue –skeletal, smooth, cardiac, contraction, muscle fibre regulation, Osseous system -structure, composition and functions of the skeleton, physiological properties of skeletal muscles and their disorders such as Rheumatoid arthritis, Gout etc. [10 Lectures]
- **Digestive and renal System**
Gross anatomy of the gastro-intestinal tract, functions of its different parts, various gastrointestinal secretions and their role in the absorption and digestion of food, peptic ulcer, ulcerative colitis, hepatic disorder. the renal system structure –Anatomy and physiology kidney; structure of the glomerulus, nephron and network of blood capillaries urinary tract, formation of urine, concentration of urine; regulation of acid-base balance; the chemical acid-base buffer systems of body fluids and Micturition, diuretics and kidney disease. [10 Lectures]
- **Respiratory system**
Anatomy of lungs, respiratory tract, mechanism and dynamics of respiration, lung volumes, transport of oxygen and carbon dioxide, disorders like cyanosis, Gas transport between the Lungs and tissues. Regulation of respiration. Respiratory adjustments in health and diseases. [4 Lectures]
- **Nervous System**
Anatomy and physiology of brain, blood-brain barrier, spinal cord, structure and types of the neuron, synapses neurotransmitters, organization of spinal and cranial nerves, central and peripheral nervous system, autonomic nervous system, receptors membrane potentials –graded potentials and action potentials, physiology of vision, audition, olfaction, taste and skin. [6 Lectures]

Text Books

1. Guyton, A.C. and Hall, J.E, **Textbook of Medical Physiology**, 13th edition, Saunders, 2015.
2. Ganong, W.F., **Review of Medical Physiology**, 26th edition, (A Lange Medical book series) McGraw –Hill (International Ed.) 2010.

References

1. Waugh, Anne and Allison Grant, **Ross and Wilson Anatomy and Physiology in Health and Illness**, 12th edition, Churchill –Livingstone / Elsevier, 2014.
2. Carola, R., J.P. Harley and C.R. Noback, **Human Anatomy & Physiology**, 2nd edition, McGraw –Hill, 1992.
3. Vander, A.J., J.H. Sherman and D.S. Luciano, **Human Physiology: The Mechanisms of Body Function**, 5th Edition, McGraw –Hill, 1990.
4. hurana, Indu A **Textbook of Medical Physiology** 2nd edition Elsevier, 2015.
5. Johnson, L.R. **Essential Medical Physiology**, 3rd Edition, Academic Press / Elsevier, 2003.